Ajay Kumar

Ph.D. Candidate in Computer Science, University of Missouri-Columbia, USA
ajay.ducs@gmail.com · ajay.kumar@missouri.edu · +1-573-801-5748
ajay-kumar.com · linkedin.com/in/xenificity · github.com/xenificity · Google Scholar · ORCID 0000-0002-9722-5649

Profile

Computer scientist who builds machine learning and data systems, then tests whether they hold up in real use. My work spans agentic AI validation at Siemens, GPU and cluster performance for genomics and neuroimaging, and crop genomics in India. I have supported researchers with production platforms, co-invented a patent-pending method, and shipped my own software.

Research Interests

Evaluation and validation of LLM and agentic systems; trustworthy and deployable machine learning; distributed and GPU computing; energy- and cost-aware scientific workflows; knowledge graphs; AI for health and science.

Education

Ph.D. in Computer Science *Expected Spring 2027*
University of Missouri-Columbia, USA. GPA 4.0/4.0. Advisor: Prof. Praveen Rao, Scalable Data Science Lab.
Dissertation: *Optimizing Scientific Workflows on NSF-Funded Cyberinfrastructure*.

M.Sc. in Computer Science *Jun 2018 – Jul 2020*
Banaras Hindu University, Varanasi, India. CGPA 8.23/10.

B.Sc. in Computer Science *2015 – 2018*
University of Delhi, New Delhi, India.

Research and Industry Experience

Research Intern, Agentic AI *Jul 2026 – Present*
Siemens Technology, Foundational Technologies, Princeton, NJ, USA. Worked with Dr. Christof Budnik.

- Led applied research on validating a production LLM-to-knowledge-graph pipeline for building and HVAC maintenance data, from an open-ended product need through to a working validation framework.
- Inspected every stage boundary of a full production run, identified eight types of silent failure, and added eight new checks to the pipeline. Three of the failure types came from deterministic code, not from the LLM.
- Outcomes: two invention disclosures, one US provisional patent application as co-inventor, and a first-author paper under review at CAIN 2027.

Graduate Research Assistant *Aug 2022 – Present*
Scalable Data Science Lab (Prof. Praveen Rao), University of Missouri-Columbia; previously in Dr. Trupti Joshi's lab.

- **DeepNeuroBench (IEEE IISWC 2026, first author).** Evaluated the DeepPrep neuroimaging pipeline on 6 public datasets and 4 cluster configurations on NSF FABRIC. The CPU-GPU cluster cut energy by about 72% and power by about 74% compared with a high-core CPU cluster. The ratio of structural to functional scan size indicated the best cluster for a dataset.
- **GPU variant calling (ACM BCB Companion 2025, first author).** Combined machine learning runtime prediction with job-shop scheduling for human genome pipelines on GPU machines. This achieved an average 2× speedup over a greedy approach and 1.6× over a dynamic approach.
- **VariantKG (Frontiers in Big Data 2025).** Helped build a knowledge graph platform that turned 1,508 human genome sequences into about 4 billion RDF statements, with graph machine learning for downstream inference.
- **Scalable variant calling (ACM CIKM 2024; NSF RAPID project).** Co-developed open-source tooling that runs variant calling on commodity clusters (github.com/MU-Data-Science/GAF).
- **G2PDeep-v2.** Built and deployed the web platform for multi-omics phenotype prediction and biomarker discovery, used by about 120 researchers since 2023 (g2pdeep.org). Presented its tutorial at ISMB/ECCB 2023.
- **NLP at scale.** Fine-tuned and benchmarked BERT and RoBERTa for sentiment analysis of about 22 million COVID-19 vaccine tweets.

Project Associate I, Statistical Genetics *Aug 2020 – Jul 2022*
International Rice Research Institute (IRRI), South Asia Hub, Hyderabad, India.

- Built **DeepMap**, an open-source deep learning package for genomic prediction in crop breeding that outperformed classical models by 13–20% and was used in about 10 later studies.
- Led a partnership between Amazon Web Services and IRRI that expanded compute capacity for genomics researchers by 50%.
- Trained more than 100 scientists and faculty in R, GWAS, QTL-seq, genomic selection, Python automation and machine learning, and worked with breeders, pathologists and molecular biologists on applied analyses.

Software Engineering Intern *Jun – Aug 2019*
School of Planning and Architecture, New Delhi, India.

- Shipped features for *Urban STeP*, a public participatory planning toolkit that reached about 15,000 visitors, and built photo tagging and the data layer for *Biodiversity of City*, an Android app, working with ecologists on the data model.

Software and Projects

- **Research systems used by others:** G2PDeep-v2 (platform, about 120 researchers), DeepMap (Python package, about 10 later studies), GAF (open-source variant calling tooling) and CyNeuro (open cyberinfrastructure course materials).
- **Independent products:** Operation Fit, a fitness planning iOS app on the Apple App Store (2026); CoMo Events, a free map of public events in Columbia, Missouri, fed by an automated calendar aggregator (2026); and StackDrop, an iOS game (2026).
- **Clinical prototype:** Equine Lameness Predictor, built with Dr. Kevin Keegan, College of Veterinary Medicine, which won first place at the Claude Builder Club Hackathon (2026). I am now re-evaluating it with leakage-free, case-level splits to learn what would hold up in practice.

Publications

★ marks five key papers. All peer-reviewed unless stated otherwise.

1. ★ **A. Kumar**, V. Omelyusik, S. Nair, P. Rao. "A Benchmark for Cost and Energy-Efficient Execution of Neuroimaging Workflows on Commodity Clusters." *IEEE International Symposium on Workload Characterization (IISWC)*, Boulder, CO, 2026. doi:10.1109/IISWC71020.2026.00042; code.
2. ★ **A. Kumar** and Siemens Technology co-authors. Paper on validating a production LLM-to-knowledge-graph pipeline. *Under review*, CAIN 2027 industry track, co-located with ICSE.
3. ★ **A. Kumar**, P. Rao, P. Sanders. "Optimizing the Variant Calling Pipeline Execution on Human Genomes Using GPU-Enabled Machines." *16th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (BCB Companion)*, Philadelphia, PA, 2025. doi:10.1145/3768322.3769016.
4. ★ S. Prasanna, **A. Kumar**, D. Rao, E. J. Simoes, P. Rao. "A Scalable Tool for Analyzing Genomic Variants of Humans Using Knowledge Graphs and Graph Machine Learning." *Frontiers in Big Data*, vol. 7, 2025. doi:10.3389/fdata.2024.1466391.
5. K. Shehzad, **A. Kumar**, M. Schutz, C. Webb, P. Nalela, M. J. Das, P. Rao. "A Scalable Tool for Democratizing Variant Calling on Human Genomes Using Commodity Clusters." *33rd ACM International Conference on Information and Knowledge Management (CIKM)*, demo track, Boise, ID, 2024, pp. 5275–5279. doi:10.1145/3627673.3679221.
6. N. Gnanapragasam, V. V. Prasanth, K. T. Sundaram, **A. Kumar**, B. Pahi, et al. "Extreme Trait GWAS (Et-GWAS): Unraveling Rare Variants in the 3,000 Rice Genome." *Life Science Alliance*, vol. 7(3), e202302352, 2024. doi:10.26508/lsa.202302352.
7. ★ **A. Kumar**, K. T. Sundaram, N. Gnanapragasam, U. M. Singh, K. J. Pranesh, et al. "DeepMap: A Deep Learning-Based Model with a Four-Line Code for Prediction-Based Breeding in Crops." *bioRxiv preprint*, not peer-reviewed, 2023. doi:10.1101/2023.07.26.550275.

Patents and Invention Disclosures

- US provisional patent application, filed October 2026, on validation of multi-stage generative AI pipelines. Co-inventor. Assigned to Siemens Corporation.
- Two invention disclosures with the Siemens Technology research team in Princeton, NJ, 2026.

Honors and Awards

- **UPE Executive Council Award** ($2,500), Upsilon Pi Epsilon International Honor Society, Dec 2025; presented as a top recipient at the 2026 UPE Annual Convention.
- **FABRIC Travel Fellowship** ($5,500), 2024. **KNIT9 / MERIF Conference Travel Award** ($2,000), 2024. **IEEE IISWC Travel Award** ($1,500), 2026. **GPC Conference Presentation Travel Award** (up to $400), University of Missouri, 2026.
- **First Place**, Claude Builder Club Hackathon, University of Missouri, 2026. **Outstanding Teaching Assistant Award**, Department of EECS, University of Missouri, 2026. **Outstanding Ph.D. Student Award**, University of Missouri. **UPE Outstanding Chapter Award** ($500), to the Missouri chapter during my term as Vice President.

Teaching, Training and Mentoring

- **Graduate Teaching Assistant**, University of Missouri: CS 4080 Parallel Programming for HPC; CS 3910 Advanced Cybersecurity; CMP_SC 1300 Computing with Data in Python; CS/ECE 4001 and CMP_SC 7001/8001 Integrated Cyberinfrastructure; CS 8001 teaching materials on cloud computing and Python.
- **Co-organizer and instructor**, NSF workshop "Scientific Workflows for Brain and Behavioral Research" (OAC-2417875), August 13–14, 2026, with 81 registrants and hands-on FABRIC sessions.
- **Curriculum developer**, NSF CyberTraining program on cyberinfrastructure for neuroscience, 2024 – Present. **Stu-**

dent mentor for project teams across all four University of Missouri System campuses, 2025 – 2026.

Invited Talks and Presentations

- Paper talks: IEEE IISWC, Boulder, CO, 2026; ACM BCB, Philadelphia, PA, 2025. Demo: ACM CIKM, Boise, ID, 2024. Tutorial: G2PDeep-v2, ISMB/ECCB, Lyon, France, 2023.
- Invited speaker on my Siemens research internship, EECS Graduate Student Association, University of Missouri, October 23, 2026. Scholarship recipient presentation, 2026 UPE Annual Convention, March 27, 2026. Poster on this benchmark work, Research and Creative Activities Forum, University of Missouri, October 28, 2026.

Leadership and Community

- **Vice President**, Upsilon Pi Epsilon, University of Missouri chapter, Aug 2024 – Present. Re-elected for 2025–26. Started a research showcase series, brought in industry speakers from NVIDIA and Netflix, and organized resume reviews with hiring managers and joint events with other student groups.
- **AI Event Ambassador and podcast host**, DeepLearning.AI, Feb 2021 – Jan 2022. Hosted 10+ episodes with AI researchers and faculty, reaching about 10,000 listeners, and served a community of 8,000+ learners.
- **Community Lead**, Community AI Hub: free AI education for learners worldwide, with a focus on reaching women. **Volunteer**, India Redefined, a non-profit. **Student Involvement Ambassador**, University of Missouri.

Professional Service

- **Reviewer**: AMIA Annual Symposium (2024; 24 reviews), NeurIPS (ethics review), ACM ICMR, Frontiers in Artificial Intelligence and other Frontiers journals. ACM Certified Reviewer, 2026. **Sub-reviewer**: IEEE MDM 2024, ACM/IEEE JCDL 2024, ICWE 2024, IEEE BIBM 2023.
- **Artifact Evaluation Committee**: ACM/IEEE CGO 2027.
- **Judge**: Mizzou Engineering High School Research Competition (2026); Show Me Research Week poster session (2026); TigerX Hackathon, University of Missouri (2025). **Technical mentor**: Entrepreneur Quest student startup competition, University of Missouri (Fall 2026).

Media Coverage

- "Computer science students take on neuroscience," Mizzou Engineering News, February 16, 2026. "Increasing access to cybertraining technology," Mizzou Engineering News, January 16, 2025.
- Interview on the Upsilon Pi Epsilon chapter, *The Maneater*, University of Missouri student newspaper.

Technical Skills

- **Machine learning and AI:** PyTorch, TensorFlow, Keras, scikit-learn; LLMs, RAG, agentic workflows, evaluation and ablation design; transformers; graph machine learning.
- **Systems:** NVIDIA Parabricks, multi-GPU computing, Slurm, Nextflow, Docker, Kubernetes, Apache Spark, Hadoop; AWS, Google Cloud, Azure; NSF FABRIC, CloudLab, ACCESS.
- **Languages and data:** Python, R, SQL, C++, Java, JavaScript, Kotlin, Swift, Bash; RDF, SPARQL, PostgreSQL, MongoDB.
- **Domains:** genomics, neuroimaging (FreeSurfer, DeepPrep), plant breeding and GWAS, building systems (HVAC). **Languages:** English (professional), Hindi (native).

Certifications and Memberships

- **Certifications:** AWS Certified Cloud Practitioner; Genomic Data Science Specialization, Johns Hopkins University; Deep Learning, AI for Medicine and Machine Learning Specializations, DeepLearning.AI; NPTEL Deep Learning, Data Science for Engineers and Introduction to Research; Responsible Conduct of Research training (CITI).
- **Memberships:** Upsilon Pi Epsilon (lifetime member); Association for Computing Machinery; Institute of Electrical and Electronics Engineers; American Society of Human Genetics.

References

Available on request.